

From results.csv, I loaded it into the Excel spreadsheet with column **N** and **Time**

N	Time
250000	0.263989
350000	0.430739
500000	0.667362
750000	1.14477
1000000	1.86765
1500000	2.86528
2000000	4.22176
2500000	5.78097
3000000	7.12254
4000000	10.3008
5000000	13.4702
6000000	16.3789
10000000	31.9104
12500000	42.009
15000000	50.4653
17500000	60.111
20000000	72.0551
25000000	92.6461

I then insert a column $N \log N$ based on N in column 2, shown below:

N	$N \log N$	Time
250000	1349485.002	0.263989
350000	1940423.816	0.430739
500000	2849485.002	0.667362
750000	4406295.948	1.14477
1000000	6000000	1.86765
1500000	9264136.889	2.86528
2000000	12602059.99	4.22176
2500000	15994850.02	5.78097
3000000	19431363.76	7.12254
4000000	26408239.97	10.3008
5000000	33494850.02	13.4702

6000000	40668907.5	16.3789
10000000	70000000	31.9104
12500000	88711375.16	42.009
15000000	107641368.9	50.4653
17500000	126753165.9	60.111
20000000	146020599.9	72.0551
25000000	184948500.2	92.6461

Next, I conduct the linear regression with X =column 2, and Y =column 3, the resulting R^2 is 0.998191.

I also ran my program on another computer with slightly better hardware configuration, output was saved in res2.csv, with the following **N**, **Time** outputs:

N	Time
250000	0.228074
350000	0.371988
500000	0.580545
750000	0.982472
1000000	1.47257
1500000	2.60357
2000000	3.67224
2500000	4.71048
3000000	5.69687
4000000	8.51401
5000000	11.0941
6000000	13.7664
10000000	25.7523
12500000	33.8212
15000000	41.4825
17500000	48.8129
20000000	59.8524
25000000	77.497

Again I inserted the 2nd column with equation $N \log(N)$ with N from the 1st column

N	$N \log N$	Time
250000	1349485	0.228074
350000	1940424	0.371988

500000	2849485	0.580545
750000	4406296	0.982472
1000000	6000000	1.47257
1500000	9264137	2.60357
2000000	12602060	3.67224
2500000	15994850	4.71048
3000000	19431364	5.69687
4000000	26408240	8.51401
5000000	33494850	11.0941
6000000	40668908	13.7664
10000000	70000000	25.7523
12500000	88711375	33.8212
15000000	1.08E+08	41.4825
17500000	1.27E+08	48.8129
20000000	1.46E+08	59.8524
25000000	1.85E+08	77.497

Next, I conduct the linear regression with X=column 2, and Y=column 3, the resulting R^2 is 0.997078, pretty similar to the prior results.

It can thus be concluded that the running time is a linear function of $N \log(n)$ with very high R^2 scores very close to 1 (>0.99) consistently.